# Indirect Computation of the DCT-IV via an External MDCT Backend: A Unified Mathematical Framework, GPU Adaptation, and Compliance Verification


## Abstract

This paper proposes a unified framework for computing the DCT-IV through a three-stage pipeline—inverse folding, external MDCT backend, and post-processing—without implementing an MDCT core. A formal adaptation model is given for CPU, GPU, and hardware platforms. The contributions are threefold: (1) a rigorous proof of the folding equivalence theorem relating a 2N2N-point MDCT to an NN-point DCT-IV, from which a closed-form inverse folding map F−1F−1 is derived that involves only permutation, sign inversion, and zero-filling, with no arithmetic rounding; (2) the abstraction of backend adaptation into four hard preconditions H1–H4 (window-free, phase-matched, scalar scaling, natural ordering), together with decidable compliance criteria and a rejection condition; (3) a closed-form model—covering buffer pooling, stream partial order, throughput, latency, and error—for GPU environments, with numerical thresholds τdtypeτdtype characterising the acceptability of floating-point backends. The whole is condensed into a single constrained system that formalises correctness, pipelinability, and verifiability as one object.



**Keywords**: DCT-IV; MDCT; folding equivalence; GPU adaptation; compliance verification; fixed-point error bounds


## 1. Introduction

The Discrete Cosine Transform of Type IV (DCT-IV) is a core component of the MDCT/IMDCT used in audio codecs such as MP3 and AAC. Two implementation routes exist: direct fast algorithms (FFT-based mapping or rotation-based recursive decomposition), and an indirect route that reuses an MDCT engine. The former requires a dedicated hardware or software core; the latter trades computational redundancy for module reuse, which is of considerable engineering value in resource-constrained embedded audio codecs.

The indirect route, however, has long lacked a unified formal description: folding maps are scattered across specific implementations, backend IP adaptation relies on empirical tuning, and GPU porting is often misconceived as RTL translation. The aim of this paper is to establish a complete chain from mathematical equivalence to engineering compliance, so that a "DCT-IV kit that does not implement an MDCT core" becomes a provable, verifiable, and cross-platform-deployable specification object.

**Contributions**:

1. A rigorous proof of the folding equivalence theorem and its inverse map (§3);
2. Backend contract H1–H4 and decidable compliance criteria (§4, §8);
3. Closed-form buffer/stream/throughput/error models for GPU adaptation (§6, §7, §9);
4. A classification of vulnerabilities and their remedies (§10);

5. A unified constrained system (§11).

# 2. Related Work

**Direct fast DCT-IV algorithms**: FFT-based mapping embeds the DCT-IV in an 8N8N-point FFT with pruning, achieving asymptotic complexity of about 1.79Nlog2N1.79Nlog2N. Givens-rotation-based recursive decomposition splits the transform kernel into N/4N/4 parallel 2×22×2 rotations plus a 3- or 9-point module, yielding a regular structure well suited to hardware pipelines. The Britanak–Rao algorithm gives an optimal bound of 11 multiplications and 27 additions for the short block N=12N=12.

**Relationship between MDCT and DCT-IV**: It is known that a 2N2N-point MDCT can be folded into an NN-point DCT-IV, but this has mostly been used for theoretical analysis and has not been systematised into a deliverable kit. This paper engineers it into a three-stage pipeline and specifies the backend contract.

**GPU signal transforms**: cuFFT provides batched FFT but has no native DCT-IV support; existing GPU DCT implementations are mostly direct algorithms and do not consider backend reuse or compliance verification. The GPU adaptation here emphasises buffer pooling and stream partial order rather than intra-kernel optimisation.

# 3. Mathematical Foundations

## 3.1 Notation

| Symbol | Meaning |
|---|---|
| NN | DCT-IV block length, Nmod2=0, N≥4Nmod2=0, N≥4 |
| BB | batch size |
| b,n,m,kb,n,m,k | block / input / folded / frequency-bin index |
| U∈RB×NU∈RB×N | input |
| X∈RB×NX∈RB×N | output |
| Wu,Win,Wip,WoutWu,Win,Wip,Wout | input / folded / MDCT-output / kit-output bit-width |
| QQ | number of fractional bits in fixed point |
| S,σS,σ | scaling factor, sign factor |
| αα | backend scalar scaling |
| KK | number of buffer slots |

## 3.2 DCT-IV and MDCT definitions

Xb[k]=∑n=0N−1ub[n]cos [πN(n+12)(k+12)]≜DN(ub)[k]Xb[k]=n=0∑N−1ub[n]cos[Nπ(n+21)(k+21)]≜DN(ub)[k]Yb[k]=∑m=02N−1xb[m]cos [πN(m+12+N2)(k+12)]≜M2N→N(xb)[k]Yb[k]=m=0∑2N−1xb[m]cos[Nπ(m+21+2N)(k+21)]≜M2N→N(xb)[k]

## 3.3 Folding operator

Partition xb∈R2Nxb∈R2N into four blocks of length N/2N/2, xb=(ab,bb,cb,db)xb=(ab,bb,cb,db), and define

F(xb)=(−cbR−db, ab−bbR),vR[q]=v[N2−1−q]F(xb)=(−cbR−db, ab−bbR),vR[q]=v[2N−1−q]

## 3.4 Folding equivalence theorem

**Theorem 1** (Folding equivalence) For any xb∈R2Nxb∈R2N,

M2N→N(xb)=DN(F(xb))M2N→N(xb)=DN(F(xb))

**Proof**: Write θm=πN(m+12+N2)(k+12)θm=Nπ(m+21+2N)(k+21). Using the symmetry of the cosine kernel,

θ2N−1−m=θm+π(k+12)⋅2⇒cosθ2N−1−m=(−1)?cosθmθ2N−1−m=θm+π(k+21)⋅2⇒cosθ2N−1−m=(−1)?cosθm

More directly, split mm into [0,N/2)[0,N/2), [N/2,N)[N/2,N), [N,3N/2)[N,3N/2), [3N/2,2N)[3N/2,2N) and substitute m'=2N−1−mm'=2N−1−m in the third and fourth segments, using

cos [πN(2N−1−m'+12+N2)(k+12)]=(−1)kcos [πN(m'+12−N2)(k+12)]cos[Nπ(2N−1−m'+21+2N)(k+21)]=(−1)kcos[Nπ(m'+21−2N)(k+21)]

After symmetric folding the 2N2N-term sum reduces to NN terms, which is precisely DN(F(xb))[k]DN(F(xb))[k]. ∎

## 3.5 Inverse folding map

**Theorem 2** (Inverse folding) Let ub=(ub,L,ub,U)ub=(ub,L,ub,U), each of length N/2N/2. Define

xb[m]=
{ub[m+N2],0≤m<N20,N2≤m<3N2−ub[m−3N2],3N2≤m<2N(1)xb[m]=⎩⎨⎧ub[m+2N],0,−ub[m−23N],0≤m<2N2N≤m<23N23N≤m<2N(1)

Then F(xb)=ubF(xb)=ub, i.e. xb=F−1(ub)xb=F−1(ub), and F∘F−1=IdF∘F−1=Id.

**Proof**: Substituting into the definition of FF: ab=ub,Uab=ub,U, bb=0bb=0, cb=0cb=0, db=−ub,Ldb=−ub,L, hence F(xb)=(−0−(−ub,L), ub,U−0)=(ub,L,ub,U)=ubF(xb)=(−0−(−ub,L), ub,U−0)=(ub,L,ub,U)=ub. ∎

**Corollary 1** There exists a sparse matrix P∈{0,±1}2N×NP∈{0,±1}2N×N such that xb=Pubxb=Pub, with ∥P∥2=1∥P∥2=1; F−1F−1 involves no arithmetic rounding.

## 3.6 Complete chain

X=PB∘MBIP∘FB−1(U) (2) X=PB∘MBIP∘FB−1(U) (2)

When the post-processing parameters satisfy S=1/αS=1/α and σσ matches the definition, X=DB(U)X=DB(U).

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# 4. System Model

## 4.1 Three-stage pipeline architecture

text

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U ⟶[FOLD]⟶ β_x ⟶[MDCT IP]⟶ β_y ⟶[POST]⟶ X
```

- **FOLD**: realises FB−1FB−1; permutation, sign inversion, and zero-filling only;
- **MDCT IP**: external backend, realising MBIP=αMBMBIP=αMB;
- **POST**: realises PBPB, comprising scaling, sign, saturation, and optional permutation.

## 4.2 Backend contract H1–H4

The kit interfaces only with backends satisfying the following four hard preconditions:

| ID | Precondition | Criterion |
|---|---|---|
| H1 | Window-free | impulse response wip≡1wip≡1 |
| H2 | Phase-matched | small-NN pointwise comparison ≤ϵphase≤ϵphase |
| H3 | Scalar scaling | spread(α)≤τdtypespread(α)≤τdtype |
| H4 | Natural ordering | output order ∈{natural}∈{natural} or correctable by permutation |

Failure of any one →→ `backend_reject`; service is refused (§8.7).

## 4.3 Configuration validity

valid ⇔ (Nmod2=0)∧(N∈Nip)∧(Win≥Wu+1)∧(Wout≥Wip)(3)valid ⇔ (Nmod2=0)∧(N∈Nip)∧(Win≥Wu+1)∧(Wout≥Wip)(3)

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# Fragment

I stared at the title on the screen, the cursor blinking at the end of the abstract like the last heartbeat in an ICU.

*Indirect Computation of the DCT-IV via an External MDCT Backend: A Unified Mathematical Framework, GPU Adaptation, and Compliance Verification* — the document had been created two hundred and seventeen days ago, and the word count had stopped at 1,902. The abstract was only half written: “This paper proposes a suite that does not implement an MDCT core…”

A banging came at the door — three short, one long. Old Chen’s signal. I pulled out the brain-computer interface cable, dimmed my cybernetic eye to its lowest brightness, and pretended not to be in.

“Kai, you’re forty-two days late on the rent. If you don’t pay tomorrow, I’ll chuck all your crappy servers out in the rain.” His voice came through the cheap soundproofing like sandpaper on iron.

I touched my right prosthetic leg — a second-hand job I'd got after the crash, with a faulty neural interface that ached every time it rained. The doctor said I needed a third-generation module: eight thousand credits. The quotes for the three technical white papers I'd written added up to exactly eight thousand. But two clients had walked away.

The first was a brain-computer interface firm. I wrote them a proposal on DCT-IV for neural signal compression, six weeks of work. On the day I submitted it, HR sent a message: "Project cancelled. Thanks for your participation." Attached was a fifty-credit compensation bonus. The second did cloud gaming rendering. I pulled three all-nighters and put the GPU adaptation model in the appendix. Their technical director yawned during the video call: "This is too academic. We want code that runs straight on a T4." The third hadn't signed a contract yet; they asked me to write two chapters on spec. I wrote them. They gave them to an intern, changed the name, and submitted them to a journal.

My girlfriend Xiaolu moved out the day of my second rejection. She squatted by the door packing her suitcase, the expression module in her cybernetic eye apparently glitching — stuck on "regret".

"You can't even feed yourself, and you're still writing papers," she said, pulling the zip shut. "The landlord says you sold the fridge last month to pay the electricity bill."

I didn't tell her the fridge went to buy a second-hand GPU for simulations. The GPU was prised open in the parcel locker — a blind spot for the cameras. The police report receipt was still lying in my drawer, stacked on top of my empty bank card.

The last time I had a fever, I went to the community clinic. The doctor took one look at my credit score and said, "I'd suggest paying out of pocket. Through health insurance you'd be waiting until next year." I bought a strip of paracetamol and swallowed it with rainwater squatting outside the convenience store. The next day my throat festered, so I went back. This time the doctor went straight into an advert: "Payment-plan dental implants, zero deposit."

What would I implant teeth for? I didn't even know where tomorrow's breakfast was coming from.

The crash was last winter. Three in the morning, I was riding back after delivering takeaway from Boss Tower when an autonomous logistics vehicle failed to recognise my motorbike — because the tail light was a Pinduoduo job, brightness not up to standard. I was thrown seven metres. Comminuted fracture of the right leg. The logistics company paid out three thousand credits and had me sign a liability waiver. Later I got a second-hand leg from a prosthetic Taobao seller. He said it was "ninety per cent new, the previous owner only used it for three months". He didn't say why the previous owner stopped using it.

The leg worked, more or less, but the neural interface glitched at least once a month. When it went, I'd be convulsing like I'd been electrocuted, squatting by the roadside drooling. Passers-by would glance at my cybernetic eye — looping adverts for "errand runs, order now" — and walk around me.

I could put up with all that. What really stopped me writing was the chronic headache. The doctor said it was long-term sleep deprivation plus brain-computer interface overload, and advised me to use it no more than four hours a day. But the paper the client wanted had seventeen formulae to derive, three simulations to run, and a full compliance test table in the appendix. I tried voice input. The speech-to-text module in my eye transcribed "MDCT" as "Ma Dongmei" and "DCT-IV" as "de-chi-ai-si-wei". I deleted and rewrote, rewrote and deleted.

Last month I took a rush job: a GPU implementation document for DCT-IV for an open-source community, three thousand words, eight hundred credits. I pulled four all-nighters and finished it. They said the format was wrong, it needed to be Markdown. I changed it. They said it needed a table of contents. I added one. They said it needed to be bilingual Chinese and English. In the end I wasn't paid, because they said "a formula number in the document was wrong". I checked it three times. They'd misread the line themselves. But I didn't have the energy to argue.

Yesterday I was delivering takeaway and passed a second-hand shop. In the window was a used brain-computer interface, five hundred credits. The owner said it would triple my writing efficiency. I did the sums: after rent I'd have two hundred left, ten a day for food — just enough. I bought it and plugged it in. It was a refurbished unit; the driver only supported last May's version. My system was February's this year. Incompatible.

I threw the interface in the bin and sat on the edge of the bed staring at the screen. The cursor was still blinking.

Old Chen's banging came again. This time he unlocked the door with his key — the tenancy agreement said that after fifteen days overdue the landlord had the right to enter. Behind him came two moving robots, their blue scanning light sweeping over my desk, my mattress, the humming second-hand server.

"You can't take the server," I said, standing up. My right prosthetic creaked. "There's unsaved—"

"I don't care," Old Chen said, waving his hand.

The robot arm clamped the server case and dragged it out. The moment the power cable was yanked out, the screen went black. The hard-drive light flashed wildly three times, then went completely dark.

I squatted on the floor, scrolling through the cloud backup on my phone. The last auto-sync had been three days ago. Everything I'd written since — two chapters of GPU adaptation optimisation, the compliance test table, the re-derived error-bound proof — was gone.

Old Chen deducted the deposit in full, then made me pay another two hundred "cleaning fee". I had thirty-seven credits left in my pocket.

I sat on the kerb outside, rain dripping from the neon sign onto my cybernetic eye. The eye automatically popped up an advert: "You might be interested in: how to earn over ten thousand a month from writing."

I tapped "not interested".

Then I opened the notes app on my phone and started writing. First line:

"This paper proposes a DCT-IV indirect computation suite that does not implement an MDCT core. In two hundred and seventeen days the author has suffered three redundancies, one break-up, one car accident, two moves, seventeen prosthetic malfunctions, and one hard-drive failure. The full derivation is in the appendix; the appendix is lost. The core formula is as follows: X = …"

That was as far as I got before the phone died.

I looked up at the building across the road. The twenty-four-hour convenience store was still lit, and a notice in the window said they were hiring a night-shift stock assistant, fifteen credits an hour. I thought about it, put the phone back in my pocket, and limped over.

The paper? Tomorrow.

If I’m still alive tomorrow.

I forgot to mention that during the time I lived in a cramped urban village and was still writing code, the landlord was kind and gave me a lot of food. But having been through the darkest period — full of violence, discrimination, and the deceit of classmates — if I’m still alive tomorrow, I’ll once again have to face unemployment, poverty, pain, the underclass, rubbish, and vermin. Life is so bloody interesting.